Gemini Chat

Example_Scribe
PDF

L22
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Generate a Lecture Scribe for exam reference from the given PPT "L22" using the structure from the PDF "Example_Scribe".
Follow the below two sets of instructions properly to generate the Lecture Scribe :
Instructions : 1) The lecture scribe is to be generated as per the structure given in Example_Scribe 2) Use the content from L22 PPT only and do not add titles/content not covered in ppt.
Instructions for following the example scribe : 1) For List of topics covered, the lecture scribe must cover the list of all topics covered in the PPT L22. Do not add new headings/topics that are not covered in ppt 2) For Explanation of Topics Covered, the lecture scribe must cover explanation of topics covered in PPT 3) For List of Definitions and Theorems, the lecture scribe must include detailed definitions with notations/assumptions along with exact statement of definitions and theorems, detailed proof of theorem step by step (proof only if given in ppt) 4) For Important Examples, try to cover more number of examples from the PPT attached (L22) in such a way that it covers all the concepts discussed. Write entire question for the example, as given in L22 PPT and write detailed step by step solution to examples. INCLUDE EXAMPLES FROM PPT ONLY 5) For the List of Important Formulas, write the list of all the important formulas with reference to topics covered in PPT (as per example_scribe format)

Based on the provided PPT content and the structure from the example document, here is the generated Lecture Scribe for your reference:

Lecture Scribe: Probability Inequalities and Bounds

1) List of Topics Covered

1. **Chebyshev's Inequality**
    - Intuition and Key Idea
    - Proof using Markov's Inequality
2. **Running Example: Coin Flipping**
    - Setup and Binomial Distribution application
    - Comparison with Markov's Bound
3. **Real-Life Applications**
    - Exam Scores Scenario
    - Network Latency SLA

2) Explanation of Topics Covered

Chebyshev’s Inequality

Chebyshev’s Inequality is a powerful tool in probability that uses both the mean (μ) and variance ($Var(X)$) to control the deviation of a random variable from its mean. It helps bound the probability of "tail regions"—values that are far from the average. Unlike Markov's inequality, which only requires the mean and applies only to non-negative variables, Chebyshev's bound often provides a tighter estimate because it incorporates the spread (variance) of the data.

Intuition

Geometrically, it limits how much area can exist in the tails of a distribution at a distance 'a' from the mean. As the sample size or certain parameters increase, these bounds typically decrease, showing that values are more likely to stay near the mean.

3) List of Definitions and Theorems

Markov’s Inequality

Definition: For a non-negative random variable X and any $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Chebyshev’s Inequality

Theorem Statement: For a random variable X with mean μ and variance σ^2, for any $a > 0$:

$$P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$$

Detailed Step-by-Step Proof:

1. Consider the event $|X - \mu|[cite_start] \geq a$.
2. Square both sides to relate it to variance: $(X - \mu)^2 \geq a^2$.
3. Apply **Markov’s Inequality** to the non-negative random variable $(X - \mu)^2$:

$$P((X - \mu)^2 \geq a^2) \leq \frac{E[(X - \mu)^2]}{a^2}$$

4. Substitute the definition of variance $Var(X) = E[(X - \mu)^2]$:

$$P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$$

4) Important Examples

Example 1: Exam Grades with Variance

Problem: The average grade is 70% and the standard deviation is 5%. The "A" cutoff is 90%. Apply Chebyshev's inequality to obtain a tighter upper bound on the fraction of "A" students.

Solution:

- **Step 1:** Identify mean and variance.

  $\mu = 70, \sigma = 5 \Rightarrow \sigma^2 = 25.$

- **Step 2:** Define the event.
  An "A" student has $X \geq 90$. Note that $X \geq 90 \Rightarrow |X - 70| \geq 20$. We use the two-sided Chebyshev bound to cover this one-sided event.

- **Step 3:** State Chebyshev with $a = 20$.

  $P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}.$

- **Step 4:** Substitute values.

  $P(X \geq 90) \leq P(|X - 70| \geq 20) \leq \frac{25}{20^2} = \frac{25}{400}.$

- **Result:** $P(X \geq 90) \leq 0.0625$ (or $\frac{1}{16}$). This is much tighter than the bound provided by Markov's inequality (0.778).

Example 2: Network Latency SLA

Problem: A network route has average latency $E[X] = 50 \text{ ms}$ and variance $Var(X) = 16 \text{ ms}^2$. Bound the probability that a packet's latency falls **outside** the range $[30 \text{ ms}, 70 \text{ ms}]$.

Solution:

- **Step 1:** Identify given parameters.

  $\mu = 50, \sigma^2 = 16.$

- **Step 2:** Rewrite failure condition as absolute deviation.
  $X \notin [30, 70] \iff |X - 50|[cite_start] \geq 20.$

- **Step 3:** Apply Chebyshev with $a = 20$.

  $P(|X - 50| \geq 20) \leq \frac{16}{20^2} = \frac{16}{400}.$

- **Result:** $P(X \text{ is outside range}) \leq 0.04.$

5) List of Important Formulas

- **Markov's Inequality:** $P(X \geq a) \leq \frac{E[X]}{a}$
- **Chebyshev's Inequality:** $P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$
- **Binomial Mean (Fair Coin):** $\mu = \frac{n}{2}$
- **Binomial Variance (Fair Coin):** $Var(X) = \frac{n}{4}$